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Course Name: Computer Vision

Course code: ITA0601

CO1-AT2

CONCEPT MAPPING

1. Overview & Features

- Enables machines to understand and interpret visual data
- Detects, recognizes & tracks objects
- Understands Scenes & Environment
- Helps in decision making

2. Digital Image Fundamentals

- Image is a 2D array of pixels
- pixel has an intensity value
- Types: greyscale & colour images
- Resolution



Pixel Matrix

3. Image Sensing & Acquisition

- Sensors: cameras, LiDAR, radar, ultrasonic, infrared etc.
- Capture real-world visual information from the environment
- Convert light signals to electrical signals.
- Mounted around the vehicle for 360 view



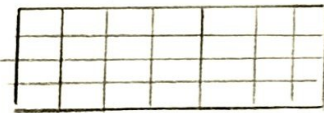
Camera



LiDAR

4. Sampling & Quantization

- Spatial Sampling: Converts a continuous image into a grid of pixels



Sampling (pixels)

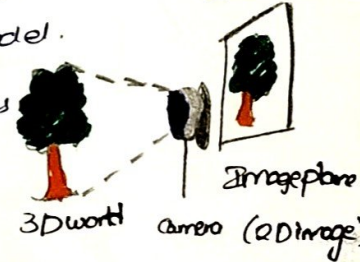
- Intensity Quantization: Converts continuous intensity values into discrete levels.



(Intensity levels)

5. Image Formation Models

- Illumination: Light from the env
- Reflection: objects reflect light to camera.
- Projection: 3D world mapped to 2D image using camera model.
- Geometric model: Relates 3D points to 2D image points (perspective projection)



7. Applications

- * Lane detection & lane keeping
- * pedestrian & cyclist detection
- * Traffic sign & signal recognition
- * Vehicle & obstacle detection
- * Navigation & path planning.
- * Self-driving decision making.

1. Overview & Features

- Helps in automatic analysis of medical images.
- Detects & recognizes anatomical structures.
- Improves accuracy & reduces human error.
- Assists doctors in treatment planning.

2. Digital Image Fundamentals

- Image is a 2D array of pixels
- Each pixel has an intensity value.
- Types: grayscale & colour images
- Resolution (Image size & clarity)

3. Image Sensing & Acquisition

- X-Rays source emits through the body.
- X-rays pass through tissues (bones absorb more, soft tissues less)
- X-ray detector
- Converts x-ray energy into electrical signals.

Medical Imaging

Applications

- * Detection of abnormalities (fractures, tumors, infections)
- * Classification of diseases (normal/abnormal)
- * Segmentation of organs & tissues
- * Measurement (size, area, volume)
- * Computer-aided diagnosis (CAD)
- * Treatment planning & monitoring.

5. Image Formation Models

- X-ray projection model: 3D body structure projected to 2D image.
- Attenuation model: x-ray intensity reduces based on tissue density.
- $I(x, y) = I_0 e^{-\mu t}$
 - * I_0 : incident x-ray intensity
 - * μ : attenuation coefficient
 - * t : thickness of tissue.
- Result: 2D grayscale x-ray image (bright = low absorption, dark = high absorption).

4. Sampling & Quantization

- Sampling: Converts the continuous x-ray image into a grid of pixels
- Determines the image resolution
- Quantization: Converts continuous intensity values into discrete gray levels
- Forms a digital image suitable for processing & analysis.